Which Account of the World?
Motivated Reasoning over Mental Representations

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Abstract

I study whether motives can affect individuals by tipping their beliefs between competing accounts of the world. I do so by proposing a theoretical model in which, in addition to distorting beliefs *within* a mental representation, individuals first choose, *among* several mental representations, the one that best trades off the motivated value of the conclusions it licenses against its defensibility. I test the model's predictions in an online experiment with a 2×2 design. The first factor varies whether the environment admits a naive but defensible mental representation alongside the rational one; the second varies whether individuals hold a directional motive. The design further separates cases in which the conclusions licensed by the naive representation are aligned with the motive from those in which they are opposed to it. When the conclusions licensed by the naive representation are aligned with the motive, I find suggestive evidence that individuals adopt it more often: the naive representation offers a convenient justification for the beliefs they favour. When instead it is the rational representation that licenses the favoured conclusions, I find no evidence that the motive moves individuals towards it. Taken together, these results suggest that motives affect the choice of mental representation asymmetrically, and only for the worse: they push individuals into a bias when the bias is convenient, but do not pull them out of one when the truth is convenient. Biases are therefore likely to be most severe precisely when they are self-serving, and individuals' own motives cannot be relied upon to debias them.

Keywords: Motivated reasoning, mental models, interaction, cognitive biases, motivated errors, correlation neglect.

1 Introduction

Consider an agent scrolling through their social media and reading personalised news about the performance of their favourite political party. Classical model of motivated reasoning suggests that the agent might read critical articals with a more skeptical eye (Taber and Lodge, 2006). However, there are also two different lenses, or "mental representations", through which the agent might understand the news feed as a whole: one that treats the information as a representative sample of opinions, and another that recognises the news as correlated and selected to fit their taste (Enke and Zimmermann, 2019; Enke, 2020). Can the agent's motive to uphold a positive image of their party affect which mental representation they engage in?

Despite large separate literatures on motivated reasoning — which studies how individuals' preferences for a rosier view of the world affect how they process information — and on mental models — which studies why and how individuals interpret their environment through representations that depart from the rational one — little is known about whether and how motives affect which mental representation individuals adopt.

In the spirit of Kunda (1990), we address this gap by shifting the focus from whether motives can distort beliefs within a given mental representation (the rational one) to whether and how motives can affect the choice of the mental representation itself.

I start by developing a theoretical model that formalises the mechanism through which motives can affect mental representation choice: when chosing a mental representation, individuals trade off representations defensibility against the motivated value of the beliefs the mental representations licence. In settings conductive to heuristics, the model also offers some predictions which I empirically test in an online experiment: when the conclusions licenced by a heuristic are in line with the motive, the heuristic can provide individuals with a *convenient justification* for holding favorable beliefs. When instead reliance on the heuristic results in undesirable conclusions, the motive can help *debiasing* individuals.

In the theoretical model, motivated reasoning operates at two stages: individuals first choose which of several mental representations to adopt, and only then form a belief within it. In the spirit of Caplin and Leahy (2001), the choice among mental representations is made by trading off the motivated value of the conclusions a representation delivers against its defensibility.

Beyond the continuous distortion within a mental representation, current models of motivated reasoning already explain, the model predicts that motivated reasoning can also produce a discrete distortion, by changing which mental representation the individual adopts. For an individual who finds several accounts defensible, the mental representation they end up adopting depends on which of the licenced conclusion the motive happens to favour.

To empirically test these predictions, an experimental setting must provide *(i)* a motive which can be exogenously introduced without altering material incentives; *(ii)* control over the set of

beliefs the motive favors; *(iii)* an environment with several defensible mental representations; and *(iv)* control over the defensibility of the different mental representations.

Environments conductive to heuristics are ideal settings to empirically test the predictions of the theoretical model. This is because, even absent a motive, individuals are faced with several defensibile mental representations and are uncertain about which one is the correct one (Bordalo et al., 2023; Enke et al., 2023). Additionally, it is possible to alter the setting to manipulate the defensibility of the different mental representations.

I use one such environment to test the theoretical predictions in an online experiment on Prolific (n = 1,212).[1] The experiment features a 2×2 factorial design, where the two factors are the possibility to engage in a heuristic and the presence of a directional motive. Following Exley and Kessler (2024), who recently highlighted the importance of understanding how biases arising from motivated reasoning interact with cognitive biases, this paper focuses on motivated errors and correlation neglect.[2][3][4]

Figure 1 summarises the resulting four conditions: *Control*, *Heuristic* (correlated signals, no motive), *Motive* (uncorrelated signals, motive), and *Motive*×*Heuristic* (correlated signals and motive). Since in the *Motive*×*Heuristic* cell the theoretical predictions differ depending on whether the motive favor the naive or the rational conclusion, this cell is split into two conditions: *Aligned* and *Opposed*.

Thanks to the design, I can directly test whether the motive merely operates "on top of" the heuristic, or whether it affects which mental representation individuals adopt. First, I compare the errors made in the *Aligned* condition to the sum of the errors observed separately in the *Motive* and *Heuristic* conditions. If the motive only operated on top of the heuristic, the two would coincide; errors in excess of the sum indicate that the motive leads more individuals to adopt the naive representation. Second, I compare the errors made in the *Opposed* condition to the difference of the errors observed separately in the *Motive* and *Heuristic* conditions. Errors below this difference indicate that the motive leads more individuals to adopt the rational representation, that is, that it helps to debias them.

The results suggest that, when the two forces operate in the same direction, individuals commit larger errors in the *Aligned* condition than they do in the *Motive* and *Heuristic* conditions combined. In other words, I find evidence of compounding, significant at the 10% level. This result is consistent with individuals switching away from Bayesian reasoning and engaging in the heuristic, but also with

[1] Prolific is [... description ...]

[2] Correlation neglect is individuals' tendency to neglect the double-counting problem arising when updating beliefs in the presence of a correlated information structure (Enke and Zimmermann, 2019).

[3] Motivated errors are errors that individuals commit when doing so allows them to justify selfish behaviour (Exley and Kessler, 2024).

[4] I focus on these biases since Exley and Kessler (2024) already provided evidence that "selfish motives can either exacerbate or mitigate errors relating to correlation neglect", making this an ideal setting to investigate the nature of the relationship between the two types of bias. The approach developed in this paper can be easily adapted to investigate the interplay between different cognitive biases and different biases deriving from motivated reasoning.

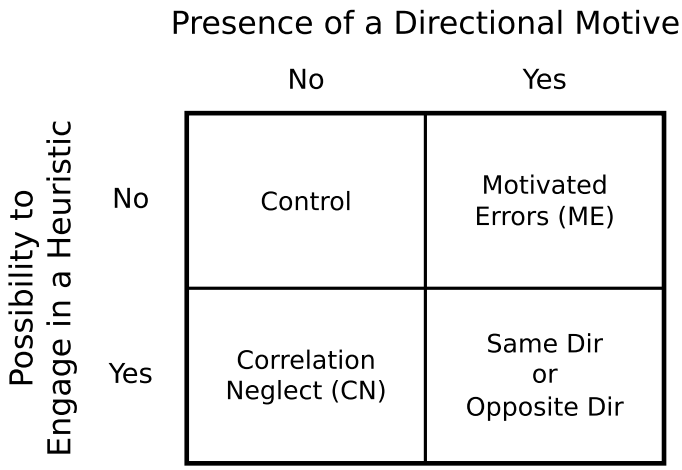


Figure 1: The 2×2 factorial design.

the idea that the naive mental model leaves individuals more vulnerable to motivated reasoning.

However, the results also suggest that individuals do not commit smaller errors in the *Opposed* condition than the difference between the errors observed separately in the *Motive* and *Heuristic* conditions. In other words, I find no evidence that the motive helps to debias individuals when the two biases operate in opposite directions. A possible interpretation for this asymmetric result is that individuals who are uncertain between the two mental models tend to reason in a Bayesian way absent the motive, and can therefore only be moved away from Bayesian reasoning once the motive is introduced.

I then examine whether these aggregate figures mask a different type of interaction than the one theorised. I start by looking at the absolute errors, which measure the size of the errors regardless of their direction. Contrary to the theoretical predictions, I find that absolute errors are larger in *Opposed* than in *Heuristic*. The difference is significant at the 5% level in the pre-registered specification, but not in the pre-registered robustness checks that exclude outliers. The increase does not come from individuals who reason rationally or naively, whose errors are of similar size in the two conditions. It comes from a small group of individuals committing large errors, both in the direction favoured by the motive and in the opposite one. A countervailing motive therefore does not debias individuals; if anything, it is associated with more large errors. Since part of these errors go against the motive, motivated reasoning alone cannot account for them: the pattern is also consistent with individuals being confused when the two forces conflict.

Finally, I classify respondents based on their errors as *rational* or as suffering from *correlation neglect*, or committing other types of errors. I then estimate a multinomial logit for the probability

of falling into each class as a function of treatment allocation. My theoretical model predicts that individuals in the *Aligned* condition are more likely to be classified as suffering from correlation neglect than individuals in the *Heuristic* condition. It also predicts that individuals in the *Opposed* condition are more likely to be classified as rational than those in the *Heuristic* condition.

I find no evidence for either relationship. This null result is itself informative. If individuals in *Aligned* are no more likely to be classified as suffering from correlation neglect, then the compounding I detect above is not coming from more people switching into the naive account. Rather, the compounding may operate through the continuous channel, with individuals committing larger motivated errors on top of whichever account they would have used anyway. This interpretation is consistent with the environments conducive to cognitive bias also being more complex and less certain (Enke et al., 2023). It suggests that it may be the added uncertainty, rather than the availability of a more plausible heuristic account, that invites more motivated reasoning, in line with Engelmann et al. (2019).

Taken together, these results paint a mixed picture of how the two forces interact. When correlation neglect and the motive point in the same direction, I find suggestive evidence that they compound. This compounding does not seem to come from individuals using the heuristic as a convenient justification for reaching the desired conclusion, but rather from the increased uncertainty associated with a more complex environment. When correlation neglect and the motive point in opposite directions, however, the picture is not one of debiasing: a countervailing motive does not pull individuals back towards rationality, nor does it make individuals with a naive mental model commit larger motivated errors; if anything, it is associated with larger absolute errors. Overall, cognitive biases and motivated reasoning appear to interact, but the interaction is direction-dependent: the motive compounds the bias when the two are aligned, and fails to correct it when they are opposed.

This paper builds on and contributes to several strands of literature. First, it bridges the literature on motivated reasoning (Kunda, 1990; Bénabou and Tirole, 2016) and on mental models (Spiegler, 2020; Bordalo et al., 2023; Gagnon-Bartsch et al., 2018).

The literature on motivated reasoning has proposed and documented several mechanisms through which directional motives can distort beliefs, including selective exposure to information (Oster et al., 2013; Ganguly and Tasoff, 2017), selective scrutiny of information (Sanitioso and Kunda, 1991; Ditto and Lopez, 1992; Taber and Lodge, 2006), and selective memory (Amelio and Zimmermann, 2023; Zimmermann, 2020). Closest to this paper are the works that look at selective aggregation of information (Eil and Rao, 2011; Möbius et al., 2022; Kahan et al., 2017). Individuals behave in a Bayesian manner absent a motive, but when a motive is introduced, they act naively. These findings point to the idea that motives do not just shape how far a given belief gets pushed once a heuristic is fixed, but also affect which reasoning structure is employed. I contribute to this literature by formalising a model of motivated selection of mental models and test its prediction

empirically.

The literature on mental models is based on the idea that individuals do not form beliefs from a unified reading of how the world works, but rather hold subjective mental models of the world which often depart from how it actually works (Esponda and Pouzo, 2016; Bohren and Hauser, 2025). Formalisations of this idea differ in where they locate the source of the mental models. It could be, for example, a misspecified causal model of how the relevant variables relate (see, e.g., Ambuehl et al., 2026; Spiegler, 2020, 2016; Schwartzstein and Sunderam, 2021), or attention to only some of the problem's features (see, e.g., Bordalo et al., 2023; Gagnon-Bartsch et al., 2018). In this paper, I contribute to this strand of literature by introducing a top-down mechanism for how individuals select which mental model to use: motivated reasoning. In this regard, my work is closest in spirit to Eliaz and Spiegler (2020), who also model individuals who "hedonically" select which narrative to adopt. In their model, individuals' choice over different narratives is constrained by objective long-run data. In the spirit of Bordalo et al. (2023), I deviate from Eliaz and Spiegler (2020) by detaching the defensibility of a mental model from objective data: in my framework, an individual can adopt a defensible mental model even if there is no data to support it. Because Eliaz and Spiegler (2020)'s narratives must fit long-run frequency data, their framework is inherently backward-looking: it speaks to recurring problems on which a track record has accumulated. Detaching defensibility from data lets my model apply equally to new situations, where no such record yet exists but a range of mental models can still be credible.

Finally, on the empirical side, this paper builds on and contributes to the small strand of literature that studies the interaction between motivated reasoning and cognitive biases. To the best of my knowledge, the only studies in this field are Ginossar and Trope (1987) and Exley and Kessler (2024). Ginossar and Trope (1987) show that individuals asked to role-play as a defence lawyer are more likely to attend to base-rate information in the classic taxi-cab problem (Tversky et al., 1982) than individuals with a pure accuracy motive — that is, a directional motive can pull people towards the normatively correct answer when it happens to align with it. Exley and Kessler (2024) show that a financial motive to over- or under-estimate can either exacerbate or mitigate errors driven by correlation neglect and anchoring, depending on whether the motive and the cognitive bias are aligned. While these studies show the robustness of motivated reasoning to settings conducive to cognitive biases, their designs and analyses do not allow the identification of a possible interaction effect. The pattern of exacerbation and mitigation they document is equally consistent with individuals experiencing a cognitive bias and layering a separate, unrelated motivated distortion on top of it, or with the two biases simply affecting different people. My design is explicitly built to directly assess the difference between the independent and joint effects of the two types of bias on individuals.[5]

[5]Additionally, in the Exley and Kessler (2024) design, the same/opposite direction condition is varied within individuals. The same participant is exposed to a motive pointing in the same direction as the bias in some rounds

The remainder of the paper proceeds as follows. Section 2 lays out the theoretical model and derives testable predictions. Section 3 describes the experimental design. Section 4 presents the empirical strategy and results. Section 5 concludes.

2 A Two-Stage Model of Motivated Reasoning

This section formalises the two-stage theoretical model sketched in the introduction. Faced with a body of evidence, the decision maker (DM) does not read it in a single, fixed way: they first select a defensible *mental model*, and only then form a belief within it. Both steps are open to motivated reasoning and I model them both in the spirit of Caplin and Leahy (2001). Across models, the motive tips the first-stage choice toward whichever credible reading of the evidence licenses the more congenial conclusion. What disciplines that choice is not the model's fit to the truth (of which the DM may be unaware) but how defensible the model is. Within a model, beliefs are then pulled toward what the DM would like to be true, at a cost that grows with the distance from the rational belief the model implies. The rest of this section builds these pieces in turn: it describes mental models in Section 2.1, it defines the mental model choice problem in Section 2.2, and then it presents the predictions the two stages jointly deliver in Section 2.3.

2.1 Mental models

Let $\theta \in \Theta$ denote the underlying object the DM is forming beliefs on and has evidence about.[6] I define a mental model $m \in \mathcal{M}$ as a mapping from the evidence, $e \in \mathcal{E}$, to a belief about θ, $g_{m,e}(\theta) \in \Delta(\Theta)$, $m : \mathcal{E} \to \Delta(\Theta)$. Concretely, m is a set of assumptions about the structure of the problem and therefore about how the evidence e is generated from θ. $g_{m,e}(\theta) \in \Delta(\Theta)$ is the posterior obtained by updating a fixed prior on θ through the likelihood those assumptions imply. The belief $g_{m,e}$ is therefore Bayesian only *internally*, relative to the structure the model assumes: it is the correct posterior given m, but coincides with the true Bayesian posterior only when m's assumptions are themselves correct.

For example, given four news reports about the performance of a political party ($e = \{\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \hat{\theta}_4\}$), one mental model, m_{iid}, might assume the four reports are iid unbiased signals of θ. Updating under this assumption weights each report equally. Another mental model, m_{bias}, might assume the reports are all inflated by a common amount. Updating under this mental model subtracts

and in the opposite direction in others. Taking the observed pattern of exacerbation and mitigation as evidence of a genuine interaction would require the same individual to be "biased" in a round, and to shake it off in the next one, only to fall for it again afterwards. My between-subject design avoids this by holding each individual's alignment fixed throughout.

[6] θ can be any object the DM might form a belief about: a point estimate, a probability distribution, or richer structure still. For finite (or countable) Θ, nothing below depends on what kind of object it is; when Θ is a continuum, the same results hold but require a calculus-of-variations argument in place of the coordinate-wise first-order condition used here. [Continuum-case proof to be added in an appendix.]

a constant from each report before aggregating. Each mental model is thus a different assumed data-generating process, and g_m is the belief a Bayesian would hold if that process were the true one.

While I do not commit to a single micro-foundation for how mental models work, an illustrative interpretation is that they are akin to *attention profiles* in the sense of Bordalo et al. (2023). There, a problem is described by a vector of F features, and a decision maker forms beliefs by encoding only some of them. An *attention profile* is a binary vector $\alpha \in \{0,1\}^F$, with $\alpha_j = 1$ if feature j is attended to and $\alpha_j = 0$ if it is neglected. Given α, the DM assigns each candidate hypothesis about θ a probability using only the statistics carried by the attended features, and then normalises across hypotheses; the resulting belief is internally coherent given α but coincides with the Bayesian posterior only when the attended features are the ones actually relevant to θ. On this reading a mental model m is an attention profile α_m, and g_m is the belief it induces. In the running example, m_{iid} neglects the feature recording whether the four reports share a common source of inflation and so treats them as independent unbiased signals, whereas m_{bias} attends to that feature and nets it out before aggregating. I return to this representation in Section 2.2.4, where a model's defensibility can be identified in terms of its salience.

In what follows, the evidence e is held fixed throughout the DM's problem, so I suppress it from the notation and write g_m for $g_{m,e}$.

2.2 Mental model choice

Having defined a mental model m and the Bayesian belief g_m it implies, I now turn to the first stage of the decision problem: the choice of the mental model. I formalise this by having the DM trade off the motivated value of the belief adopted given the mental model, $V(m)$, against the defensibility of the mental model, $D(m)$. In this stage, the DM chooses a mental model m from the set of available mental models $\mathcal{M}$ to maximize the sum of these two objects, weighted by a malleability parameter λ.

$$m^* \in \arg\max_{m \in \mathcal{M}} \; V(m) + \frac{1}{\lambda} D(m) \tag{1}$$

Adopting a less defensible mental model is worthwhile only when the belief it licenses delivers enough additional motivated value to offset the defensibility conceded.

The remainder of this section unpacks the two objects that (1) trades off: the indirect motivated value of a mental model, $V(m)$, and its defensibility, $D(m)$.

2.2.1 The indirect value of a mental model: $V(m)$

In my framework, the DM does not gain utility directly from holding a mental model, m, but rather from the belief $\gamma \in \Delta(\Theta)$ it licenses. In line with the motivated-reasoning literature, while

the rational belief implied by a mental model, g_m, is fixed, I allow the DM to still manipulate their belief *within* a mental model.

Following Caplin and Leahy (2001), I assume that the DM gains utility from holding a belief γ under the mental model m according to:

$$U(\gamma, m) = v(\gamma) - \frac{1}{\lambda}\,\mathrm{KL}(\gamma|g_m), \qquad \mathrm{KL}(\gamma|g_m) = \sum_{\theta} \gamma(\theta) \ln \frac{\gamma(\theta)}{g_m(\theta)} \tag{2}$$

In equation (2), $v(\gamma)$ represents the psychological payoff the DM gains from holding beliefs that assign higher probabilities to outcomes that are favorable to them. In a sense, $v(\gamma)$ ranks beliefs by how gratifying they are to hold (rather than by their accuracy). For instance, if θ is the performance of the DM's favourite political party, v is higher for beliefs γ that place more probability mass on higher values of θ: the DM enjoys thinking of their party as a successful one.

The second term in equation (2) captures the mental cost for departing from the rational belief g_m implied by the mental model m. This cost is given by the Kullback–Leibler divergence between γ and g_m,[7] scaled by a malleability parameter λ that governs how costly it is to believe something other than g_m.

Because the DM can still spin their belief within the model, the value of holding m is the utility of that optimally spun belief, not the utility of g_m itself:

$$\gamma_m^* \in \arg\max_{\gamma \in \Delta(\Theta)} U(\gamma, m), \qquad V(m) = U\big(\gamma_m^*, m\big). \tag{3}$$

Evaluating $V(m)$ therefore requires solving the within-model belief choice first — the *second stage* of the problem — which I do in the next subsection, before returning to the first-stage choice of m.

2.2.2 Belief choice within a mental model (second stage)

In this subsection, I describe how the DM chooses γ to maximize the utility function described in (2), over $\gamma \in \Delta(\Theta)$.

For a tractable, closed-form solution that allows examining comparative statics, I impose two restrictions. First, let v be locally linear around g_m: $\partial v / \partial \gamma(\theta) \approx \psi(\theta)$. $\psi(\theta)$ is independent of γ and offers the state-by-state marginal reward of the motive. This allows me to write $v(\gamma) \approx v(g_m) + \sum_\theta \psi(\theta)\big(\gamma(\theta) - g_m(\theta)\big)$.[8] Second, approximate the divergence $\mathrm{KL}(\gamma|g_m)$ by its second-order Taylor expansion around $\gamma = g_m$. Define the spin $x(\theta) := \gamma(\theta) - g_m(\theta)$, so that $\sum_\theta x(\theta) = 0$ since

[7] An integral against θ in place of the sum when Θ is a continuum.

[8] This is exact, not an approximation, whenever, as in Caplin and Leahy (2001), v is itself linear in γ (e.g. the anticipatory-expected-utility form $v(\gamma) = \sum_\theta \gamma(\theta) u(\theta)$ for a fixed per-state payoff u, in which case $\psi = u$). It is a genuine approximation, valid only for γ^* close to g_m, whenever v is curved in γ: any curvature of v (e.g. diminishing psychological returns to an increasingly favorable belief) is then discarded, and whatever resistance to further distortion that curvature would have provided is instead attributed entirely to the divergence cost $1/\lambda$.

both γ and g_m are probability distributions. Substituting $\gamma(\theta) = g_m(\theta)+x(\theta)$ and Taylor-expanding the log to second order in x gives

$$\begin{aligned}
\mathrm{KL}(\gamma|g_m) &= \sum_\theta \gamma(\theta) \ln \frac{\gamma(\theta)}{g_m(\theta)} \\
&= \sum_\theta \big(g_m(\theta) + x(\theta)\big) \ln\!\left(1 + \frac{x(\theta)}{g_m(\theta)}\right) \\
&\approx \sum_\theta \big(g_m(\theta) + x(\theta)\big) \left(\frac{x(\theta)}{g_m(\theta)} - \frac{1}{2}\left(\frac{x(\theta)}{g_m(\theta)}\right)^2\right) \\
&= \sum_\theta \left(x(\theta) + \frac{1}{2}\frac{x(\theta)^2}{g_m(\theta)}\right) + O(x^3) \\
&= \frac{1}{2}\sum_\theta \frac{x(\theta)^2}{g_m(\theta)} + O(x^3), \qquad (4)
\end{aligned}$$

where the last line uses $\sum_\theta x(\theta) = 0$ to cancel the first-order term. Dropping the $O(x^3)$ remainder leaves the χ^2-approximation $\mathrm{KL}(\gamma|g_m) \approx \frac{1}{2}\sum_\theta x(\theta)^2/g_m(\theta)$, valid for directions satisfying $\sum_\theta x(\theta) = 0$.

This leaves me a much easier problem: substituting both approximations into (2), the DM's problem becomes the quadratic program

$$\max_x \; v(g_m) + \sum_\theta \psi(\theta)\, x(\theta) \;-\; \frac{1}{2\lambda}\sum_\theta \frac{x(\theta)^2}{g_m(\theta)} \qquad \text{s.t.} \;\; \sum_\theta x(\theta) = 0. \qquad (5)$$

The Lagrangian first-order condition on (5), with multiplier ν on the adding-up constraint, is

$$\begin{aligned}
&\psi(\theta) - \frac{x^*(\theta)}{\lambda g_m(\theta)} = \nu \qquad \text{for every } \theta \in \Theta \\
\iff \quad & x^*(\theta) = \lambda g_m(\theta)\big(\psi(\theta) - \nu\big) \qquad (6)
\end{aligned}$$

Summing (6) across θ and imposing $\sum_\theta x^*(\theta) = 0$ together with $\sum_\theta g_m(\theta) = 1$, I get:

$$\nu = \sum_\theta g_m(\theta)\psi(\theta) =: \mathbb{E}_{g_m}[\psi] \qquad (7)$$

Finally substituting ν back into (6) gives the optimal distortion $x^*(\theta)$, and therefore the optimal belief $\gamma^*(\theta) = g_m(\theta) + x^*(\theta)$:

$$\begin{aligned}\gamma^*(\theta) &= g_m(\theta) + x^*(\theta) \\ &= g_m(\theta) + \lambda\, g_m(\theta)\Big(\psi(\theta) - \mathbb{E}_{g_m}[\psi]\Big) \\ &= g_m(\theta) + \lambda\, \mathrm{Cov}_{g_m}\big(\mathbb{1}_{\{\theta\}},\, \psi\big) \end{aligned} \tag{8}$$

In the optimal belief, probability moves toward state θ in proportion to how much better the motive rewards θ than it rewards the average state under the reference belief g_m, $\psi(\theta) - \mathbb{E}_{g_m}[\psi]$, scaled by $g_m(\theta)$ itself: a reference belief that already puts little weight on θ has little room to inflate it further, and one certain of some state has no room to move at all. Equivalently, the belief moves in the direction of, and in proportion to, the covariance between believing θ and the motive's reward ψ.

2.2.3 From the chosen belief to the indirect value of model m

Having solved the belief-choice subproblem, I can fold its solution back into $U\big(\gamma, m\big)$ to obtain $V(m)$. Retaining the two approximations from Section 2.2.2 I obtain:

$$V(m) = v(g_m) + \sum_\theta \psi(\theta)\, x^*(\theta) \;-\; \frac{1}{2\lambda} \sum_\theta \frac{x^*(\theta)^2}{g_m(\theta)}. \tag{9}$$

The first term is the motive-assessed value of the Bayesian beliefs implied by the model: a mental model whose implied Bayesian belief already places weight where the motive wants it is more attractive to the DM before any distortion enters. The second and third are the motivated gain and the divergence cost of the optimal motivated spin: in addition to the primary motivated value, the DM gains an additional bonus for holding a mental model that also leaves room for profitable spins.

Plugging in the optimal distortion $x^*(\theta) = \lambda\, g_m(\theta)\big(\psi(\theta) - \mathbb{E}_{g_m}[\psi]\big)$ from (8), the gain term collapses to a variance:

$$\sum_\theta \psi(\theta)\, x^*(\theta) = \lambda \sum_\theta g_m(\theta)\, \psi(\theta)\big(\psi(\theta) - \mathbb{E}_{g_m}[\psi]\big) = \lambda\big(\mathbb{E}_{g_m}[\psi^2] - \mathbb{E}_{g_m}[\psi]^2\big) = \lambda\, \mathrm{Var}_{g_m}(\psi), \tag{10}$$

and the divergence cost is exactly half of it:

$$\frac{1}{2\lambda} \sum_\theta \frac{x^*(\theta)^2}{g_m(\theta)} = \frac{\lambda}{2} \sum_\theta g_m(\theta)\big(\psi(\theta) - \mathbb{E}_{g_m}[\psi]\big)^2 = \frac{\lambda}{2}\, \mathrm{Var}_{g_m}(\psi). \tag{11}$$

The DM thus keeps half the gross motivated gain, and $V(m)$ collapses to:

$$V(m) = v(g_m) + \frac{\lambda}{2} \sum_{\theta} g_m(\theta) \Big(\psi(\theta) - \mathbb{E}_{g_m}[\psi] \Big)^2$$
$$= v(g_m) + \frac{\lambda}{2} \operatorname{Var}_{g_m}(\psi). \tag{12}$$

The bonus gained in addition to the primary motivated value of having a mental model that implies a Bayesian belief g_m is increasing in $\operatorname{Var}_{g_m}(\psi)$: the more g_m spreads probability mass across states with different values of ψ, the more room the DM has to profitably spin their belief within the model.[9] At the other extreme, a mental model that places all its weight on a single state has $\operatorname{Var}_{g_m}(\psi) = 0$ and earns no bonus, since there is no room left to spin.

Formally, let g' be a mean-preserving spread of g in ψ: $\mathbb{E}_g[\psi] = \mathbb{E}_{g'}[\psi]$, but g' shifts weight toward more extreme values of ψ. Then $\operatorname{Var}_{g'}(\psi) \geq \operatorname{Var}_g(\psi)$, so $V(g') \geq V(g)$ whenever $v(g') = v(g)$: two mental models that the motive rates equally on their honest merits can still differ in how attractive they are to the DM, purely because one leaves more room for motivated spin than the other.

2.2.4 The defensibility of a mental model $D(m)$

The second object in (1), $D(m)$, measures how defensible a mental model is as a reading of the evidence. This is the constraint that disciplines the choice of model, the first-stage counterpart of the role the reference belief g_m plays in disciplining second-stage spin. Engaging a mental model the DM finds natural is almost free; talking oneself into a far-fetched one is costly, and (1) makes the DM bear that cost. A less defensible model is therefore adopted only when the belief it licenses delivers enough additional indirect motivated value $V(m)$ to offset the defensibility conceded.

Crucially, $D(m)$ is not disciplined by the mental model's fit to the truth. The DM's choice of mental model is disciplined by how defensible it is, not by how accurate it is. A mental model can thus be both cheap to adopt and inaccurate, so biased reasoning survives in my framework even when the motive is absent or when there are no observed long-run frequencies to learn from. Detaching defensibility from accuracy sets my framework apart from the narratives of Eliaz and Spiegler (2020), whose "credibility" is pinned down by consistency with observed long-run frequencies.

I do not commit to a single micro-foundation for D. One natural discipline, however, is worth spelling out, both because it is concrete and because it connects the first stage to an existing account of why people read the same evidence differently. Recall from Section 2.1 that a mental model can be represented as an attention profile $\alpha \in \{0,1\}^F$ over the problem's features, with g_α the belief that profile induces (Bordalo et al., 2023). In the Bordalo et al. (2023) framework, bottom-up

[9] Since $\operatorname{Var}_{g_m}(\psi) = \operatorname{Cov}_{g_m}(\psi, \psi)$, this bonus is the same covariance object that governs the optimal spin in (8), now instantiated with ψ against itself instead of against the indicator $\mathbb{1}_{\{\theta\}}$.

attention selects the profile that maximises total salience:

$$\alpha^* \in \arg\max_{\alpha} \; C(\alpha) + P(\alpha) + \epsilon_\alpha, \tag{13}$$

where $C(\alpha)$ is the *contrast* of the resulting representation — how sharply it favours one hypothesis over the other — $P(\alpha)$ is its *prominence* — how top-of-mind the attended features are, given how the problem is described — and ϵ_α is an individual-level salience shock.

Under this framework, the defensibility of a mental model can be considered as the salience of its attention profile, $D(\alpha) := C(\alpha) + P(\alpha) + \epsilon_\alpha$. This turns the first stage into a *motivated choice of attention profile*: the DM picks α to maximise $V(\alpha) + c\big(C(\alpha) + P(\alpha) + \epsilon_\alpha\big)$, with bottom-up salience setting the default mental model and the motive acting as a top-down override. In a non motivated setting, the DM simply picks the most salient profile, and (1) reduces to (13): my first stage nests Bordalo et al. (2023) as the no-motive case.

2.3 Predictions

The two stages together generate three testable predictions.

1. **Within-model, motivated distortion.** Holding the mental model fixed, beliefs move away from the model-implied belief g_m toward the states the motive rewards, in proportion to $\mathrm{Cov}_{g_m}\big(\mathbb{1}_{\{\theta\}},\, \psi\big)$ (as described in equation (8)). This is the Caplin and Leahy (2001) channel, recovered as my second stage.

2. **Unmotivated model choice.** Absent a motive ($V(m) = 0,\, \forall m \in \mathcal{M}$), the DM chooses the most defensible mental model (which might not be the rational one). This is the Bordalo et al. (2023) channel, recovered as my first stage absent a motive.

3. **Motivated choice of model.** When the DM finds two mental models comparably defensible ($D(m) \sim D(m')$), the motive tips the first-stage choice (described in equation (1)) toward whichever model licenses the desired belief. Behaviour then switches discretely between the accurate and the biased model with the *direction* of the motive.

3 Experimental design

To empirically test these predictions, an experimental setting must provide *(i)* a motive which can be exogenously introduced without altering material incentives; *(ii)* control over the set of beliefs the motive favors; *(iii)* an environment with several defensible mental models; and *(iv)* control over the defensibility of the different mental models.

To satisfy these requirements, I propose an experimental design that features a 2×2 factorial design, where the two factors are the presence of a directional motive and the presence of an environment conducive to a cognitive bias.

Following Exley and Kessler (2024), who recently highlighted the importance of understanding how biases arising from motivated reasoning interact with cognitive biases, this paper focuses on motivated errors and correlation neglect. Therefore, the first factor involves manipulating the possible presence of correlation neglect while the second pertains the manipulation of individuals' motives. As shown in Figure 1, participants were randomly assigned to one of four conditions: *Control*, *Heuristic*, *Motive*, and *Motive*×*Heuristic*, with the latter condition further divided into two sub-conditions, depending on whether the motive and the bias push beliefs in the same or opposite directions (*Aligned* or *Opposed*).

The experiment consists of two parts. First participants undertake the "calibration" part. In this part, I elicit a participant-specific monetary value that makes the participant indifferent between receiving that amount as a bonus payment and donating a fixed amount to a charity. This calibration is necessary to be able to manipulate the motive without altering incentives. Then, participants complete the "main" part of the experiment in which they are randomly assigned to one of the four experimental conditions and complete the main experimental tasks.

To incentivise participants to excert effort and answer truthfully, at the end of the experiment, one part was randomly selected as the "part that matters" for each participant. Participants received a bonus payment and/or a donation to a charity was made depending on their choices in that part.

3.1 Charity Equivalent Calibration

In the calibration part, I elicit a participant-specific monetary value X at which the participant is indifferent between receiving X as a bonus payment and donating \$1.50 to a designated charity (Emergency USA). Following Exley and Kessler (2024), I measure this value using a multiple price list. Each choice in the list consists of two options: (i) a certain amount of money as a personal bonus payment, or (ii) a donation to the charity. The donation amount is fixed at \$1.50, while the bonus payments ranged from \$0.00 to \$1.50 with \$0.05 increments. I incentivize truthful reporting by randomly selecting one of the choices at the end of the experiment and implementing it if the first part is randomly selected as the "part that matters".

The multiple price list generates an indifference range for the X value. For example, if a participat prefers the \$1.50 donation over a \$0.95 bonus payment (and any smaller amount), but prefers a \$1.00 bonus payment (and any larger amount) over the donation, their indifference range is $(\$0.95, \$1.00)$. For each participant I define X as the lower bound of their indifference range.[10] The

[10] Unless the lower bound of the indifference range is \$0.00, in which case I assign a value of \$0.05.

distribution of the lower bound of the indifference range is presented in Appendix Figure XXX.[11]

As mentioned above, before completing the multiple price list, participants are informed that at the end of the experiment, this part would be selected for determining the bonus payment with 50% probability, and that if selected, one of their choices would be randomly selected and implemented.[12]

3.2 Main task

Participants then proceed to the main part of the experiment, which consists of six estimation tasks. In each task, participants estimated the average of four numbers (henceforth, "items"). For example, if the items were I={18, 34, 68, 92}, then the target quantity participants had to estimate was (18+34+68+92)/4=53. Participants, however, did not observe the items directly, but received a signal for each item. The signals correlational structure and the motives participants faced depended on the thier treatment condition.

In the *Control* condition, subjects faced a simple environment without the possibility of correlation neglect. Each signal was an unbiased and precise represention of an item. For example, if the items were I={18, 34, 68, 92}, then the signals would be S={18, 34, 68, 92}. Therefore, in this condition only one possible mental model was available: the items are uncorrelated. Furthermore, in this condition, individuals had no motive affecting their estimation. Participant were eligible for a bonus payment only if they correctly estimated the average of the items in a randomly selected task. The amount of the bonus payment was set equal to the X value determined in the calibration part.

In the *Heuristic* condition, participants received correlated signals rather than the direct item values. Specifically, the first signal was exactly the highest or the lowest item value, while the remaining signals were the average of this first signal and other items For example, if the items were I={18, 34, 68, 92}, then the signals could be S={18, 26, 38, 55} or S={92, 55 63, 80}. The correlational structure of the signals was known to participants. In the example, subjects would know that the first signal $s_1 = 18$ precisely represented an item and that the signal $s_2 = 26$ was generated by averaging the first signal, s_1, and an "unknown" item ($i_2 = 34$). Note that when the first signal was high (low), the other signals were also biased upward (downward), inducing estimation errors. For this reason, I introduced two sub-conditions: in the *Heuristic+* condition, the first signal (the precise one) corresponded to the highest item, whereas in the *Heuristic-* condition, the first signal corresponded to the lowest item. These sub-conditions remained costant within individuals across tasks. In this condition, participants could engage in two defensible mental models. Rationally, they could recognise the correlated structure of the signals and keep it into account when estimating the average of the items. Naively, they could ignore the correlated structure and

[11] Results are robust to a restricted sample that excludes the Y% of participants whose lower bound is \$0.00.

[12] For example, if a participant's indifference range was (\$0.95, \$1.00), and the randomly selected choice was between a \$0.90 bonus and a \$1.50 donation, they would receive no bonus, and the donation would be made.

treat the signals as independent. Participants in the *Heuristic* condition had no motive distorting their estimates and were elegible for the X-value bonus payment only if they correctly estimated the average of the items in a randomly selected task(as in the *Control* condition).

In the *Motive* condition, participants faced a simple environment with unbiased and precise signals (as in the *Control* condition). However, in this condition, I provided respondents with a *motive* to over- or under-estimate the average of the items. Following Exley and Kessler (2024), I created this motive without altering participants' incentives by introducing an additional payment component (on top of the bonus payment paid only in case of a correct estimate). The additional payment component was paid indipendently from the participants' response, but it took a different "form" depending on the response. If the participant over- (under-)estimated the average, the additional payment was a personal payment equal to the X value determined in the first part. If the participant under-(over-)estimated or correctly estimated the average, the additional payment was a \$1.50 donation to charity. Since the X value was defined to make participants (almost) indifferent between receiving \$X personally and donating \$1.50, they should be indifferent between over- and under-estimations making the additinal payment independent of their incentives. Participants should still strictly prefer a correct estimate since on top of the independent additional payment (in the form of a charity donation), they would gain the bonus payment. While individuals' incentives should be unaffected, Exley and Kessler (2024) showed that individuals gain a motive to secure the personal payment. For this reason, I consider two sub-conditions: in the *Motive+* condition, participants secured the personal payment by overesstimating the average of the items, whereas in the *Motive-* condition, they secured the personal payment by underestimating. These sub-conditions remained costant within individuals across tasks.

Finally, in the *Motive*×*Heuristic* condition, participants received correlated signals (as in the *Heuristic* condition) and had a *motive* to over- or under-estimate the average of the items (as in the *Motive* condition). The interaction of cognitive bias and motivated reasoning could lead to different behaviors depending on whether the motive and bias aligned or opposed each other. Thus, I introduced two sub-conditions: in the *Aligned* condition, cognitive bias and motivated incentives pushed beliefs in the same direction, whereas in the *Opposed* condition, they operated in opposite directions

3.3 Procedural details

The experiment involved XXX subjects recruited from the Prolific platform. Each participant received a baseline payment of \$YYY for completing the 15-minute study. Additionally, one randomly selected decision per participant was implemented for an additional payment, which resulted in a bonus payment to the participant and/or a donation to charity. The average bonus payment was \$ZZZ and the average donation was \$WWW.

At the beginning of each part, individuals receive instructions that provide guidance on the tasks they will perform. In the main part, the instructions are tailored to the individuals' specific conditions. Therefore, individuals implicitly learn about their condition through the structure of their payoff. Participants are not informed about the existence of different conditions. Following the instructions, individuals answer comprehension questions. The correct answer to all the comprehension questions is required to proceed to the practical parts. If a participant fails to correctly answer a comprehension question twice, they are excluded from the study.

After concluding the tasks in the main part, I ask participants whether they used external resources to complete the questions other than calculators (e.g. chatGPT). While I encourage participant to answer truthfully, I cannot verify their responses. As from preregistration, I excluded participants who reported using external resources from the analyses.

Additionally, I elicit participants' cognitive uncertainty about their responses in the main task. In practice, for each decision participants took, they are asked how certain they are that the correct answer is contained in a certain window around their actual decision (in the spirit of Enke and Graeber, 2023).The elicitation of the cognitive uncertainty is postponed to the end of the studies to minimize participants' cognitive load while answering the questions and to simplify the initial instructions. This variable is employed to explore potential variations in treatment effects among participants.

Finally, participants complete a questionnaire that gathers information about demographic variables. These variables are employed to assess the effectiveness of the randomization procedure.

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A Charity Equivalent

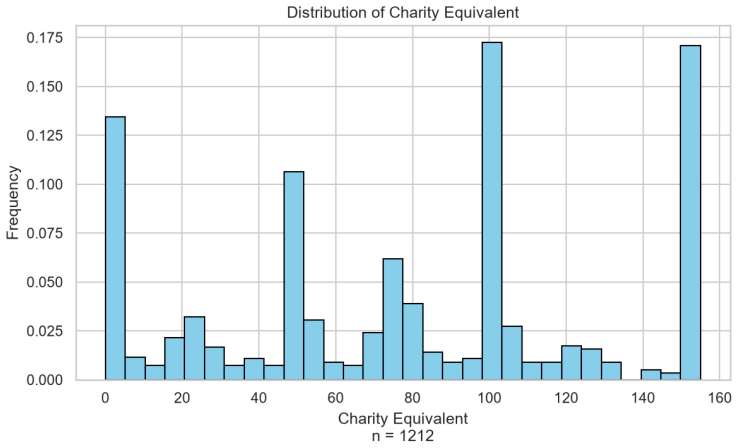


Figure 2: Distribution of elicited charity-equivalent upper bounds (ce_ub) across all subjects.